Cantor's Theorem and the Power-Set Hierarchy

An End-to-End Meta-Math Walkthrough

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Abstract

For any proposed enumeration of subsets, choose a new subset that disagrees with the assigned subset at each element's own membership test. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

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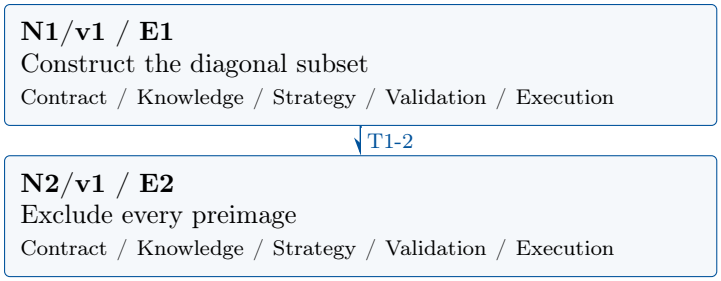
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *For every set A, prove there is no surjection $f : A \to \mathcal{P}(A)$, where $\mathcal{P}(A)$ is the set of all subsets of A. Since $a \mapsto \{a\}$ is an injection, conclude that A has strictly smaller cardinality than its power set, in the sense of injection plus absence of a bijection. The diagonal argument does not require the axiom of choice.*

A. Foundations and permitted imports

Use the power-set and separation axioms of ordinary set theory, the definitions of injective, surjective, and bijective functions, and classical propositional logic. Separation permits forming a subset of an already given set by a formula. No unrestricted "set of all sets" comprehension is used.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-2: N1/E1/v1 to N2. $D = \{a \in A : a \notin f(a)\}$ is a well-defined member of $\mathcal{P}(A)$.

Receiving activity. N2.1

Compatibility. E1/v1 and a hypothetical $b \in A$ with $f(b) = D$. Check: The contradiction applies to every proposed surjection, including finite and empty domains.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Construct the diagonal subset

Object and role. Use bounded separation to reverse each element's own membership decision.
Local obligation: $D = \{a \in A : a \notin f(a)\}$ is a well-defined member of $\mathcal{P}(A)$.

Inputs. An arbitrary function $f : A \to \mathcal{P}(A)$.

C — Contract

Publish E1/v1 only after establishing this obligation: $D = \{a \in A : a \notin f(a)\}$ is a well-defined member of $\mathcal{P}(A)$.

K — Knowledge

An arbitrary function $f : A \to \mathcal{P}(A)$. Retain the stated assumptions and import versions.

S — Strategy

Use bounded separation to reverse each element's own membership decision.

V — Validation

The definition produces a subset of A, so it belongs to the codomain; no self-containing universal set is needed.

E — Execution

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 For each $a \in A$, $f(a)$ is a subset of A, so the membership statement $a \in f(a)$ is meaningful. By separation, the elements of A for which this statement is false form a set $D \subset A$.

N1.2 The definition gives, for each $a \in A$, the exact equivalence $a \in D \iff a \notin f(a)$. In particular D is in the stated codomain $\mathcal{P}(A)$.

Export E1/v1. $D = \{a \in A : a \notin f(a)\}$ is a well-defined member of $\mathcal{P}(A)$.

Acceptance gate. The definition produces a subset of A, so it belongs to the codomain; no self-containing universal set is needed.

Targeted repair. If the construction is made over all sets rather than A, restrict it to A; bounded separation is the actual permitted axiom. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Exclude every preimage

Object and role. Specialize the membership equivalence at the alleged preimage and derive a contradiction. **Local obligation:** No $f : A \to \mathcal{P}(A)$ is surjective; the singleton map establishes strict cardinal growth.

Inputs. E1/v1 and a hypothetical $b \in A$ with $f(b) = D$. Incoming proof exports: E1/v1 at N2.1.

C — Contract

Publish E2/v1 only after establishing this obligation: No $f : A \to \mathcal{P}(A)$ is surjective; the singleton map establishes strict cardinal growth.

K — Knowledge

E1/v1 and a hypothetical $b \in A$ with $f(b) = D$. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

S — Strategy

Specialize the membership equivalence at the alleged preimage and derive a contradiction.

V — Validation

The contradiction applies to every proposed surjection, including finite and empty domains.

E — Execution

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 If $f(b) = D$, the equivalence in E1 reads $b \in D \iff b \notin D$, impossible in classical logic. Thus no element of A maps to D, and the arbitrary function f is not surjective.

N2.2 The singleton map $a \mapsto \{a\}$ is injective because equal singleton sets have equal elements. A bijection would in particular be a surjection, already ruled out. This proves the claimed strict cardinal comparison. For $A = \varnothing$, $\mathcal{P}(A) = \{\varnothing\}$ and the same lack of surjection is immediate.

Export E2/v1. No $f : A \to \mathcal{P}(A)$ is surjective; the singleton map establishes strict cardinal growth.

Acceptance gate. The contradiction applies to every proposed surjection, including finite and empty domains.

Targeted repair. If a contradiction is interpreted as showing the power set is inconsistent, identify the rejected assumption: only the proposed surjectivity fails. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Construct the diagonal subset

1. **N1.1.** For each $a \in A$, $f(a)$ is a subset of A, so the membership statement $a \in f(a)$ is meaningful. By separation, the elements of A for which this statement is false form a set $D \subset A$.

2. **N1.2.** The definition gives, for each $a \in A$, the exact equivalence $a \in D \iff a \notin f(a)$. In particular D is in the stated codomain $\mathcal{P}(A)$.

B. N2 — Exclude every preimage

1. **N2.1.** If $f(b) = D$, the equivalence in E1 reads $b \in D \iff b \notin D$, impossible in classical logic. Thus no element of A maps to D, and the arbitrary function f is not surjective.

2. **N2.2.** The singleton map $a \mapsto \{a\}$ is injective because equal singleton sets have equal elements. A bijection would in particular be a surjection, already ruled out. This proves the claimed strict cardinal comparison. For $A = \varnothing$, $\mathcal{P}(A) = \{\varnothing\}$ and the same lack of surjection is immediate.

□

V. WORKED ORCHESTRATION

For $A = \{1, 2, 3\}$, consider $f(1) = \{1, 2\}$, $f(2) = \{1\}$, and $f(3) = \{3\}$. The diagonal rule includes 2 but excludes 1 and 3, so $D = \{2\}$. It is different from all three listed images, illustrating the exact self-index disagreement.

A. Boundary and hypothesis audit

Listing all finite subsets of $\mathbb{N}$ is possible and does not contradict Cantor: the codomain is then only a proper part of $\mathcal{P}(\mathbb{N})$. The diagonal subset for such a list cannot be one of its listed finite subsets. Replacing "all subsets" by "finite subsets" changes the target.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected scope (N1): form the set of all sets that do not contain themselves. **Diagnosis:** Knowledge replaced separation over A by unrestricted comprehension. **Repair:** construct only $D \subset A$ using the given map; its existence is licensed by separation. Then N2 contradicts surjectivity, not the set axioms. Recheck the codomain membership and final cardinal comparison.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Apply the construction to $A = \mathbb{N}$ to obtain an uncountable power set and to explain why no single list can enumerate all binary sequences.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. Andrew Marks, Set Theory notes for sets and ultraproducts; Berkeley, Mathematical Logic for Mathematicians, Part I for first-order semantics; Andrew Marks, Computability Theory notes for effective procedures and undecidability. The proofs below specify their own set-theoretic and semantic assumptions.